

Raspberry Pi 3 comes with 64-bit processor and built-in Wi-Fi

The Raspberry Pi Foundation has launched the new Pi 3 with Wi-Fi and Bluetooth, but at the price of Pi 2. The new version is an iterative update on the Pi 2.

According to Raspberry Pi Foundation's CEO, Eben Upton, the Pi 3 has been in development for over a year. The new version comes with a quad-core 64-bit 1.2GHz ARM Cortex A53 chip. This is about 50 per cent faster than its predecessor. Upton further pointed out that, "Our primary goal in moving to A53 from A7 was to get a better 32-bit core. A53 running in 32-bit mode outperforms A7 at the same clock speed by 20-30 per cent."



The new Pi 3 comes with the same 1GB RAM and the clock speed has been upgraded to 900MHz from 450MHz. There is also an improvement in the VideoCore IV graphics from 250MHz to 400MHz. Both of these are based in a board that retains the same dimensions as the Pi 2. The addition of integrated Bluetooth 4.1 and 802.11n Wi-Fi will help in reducing the need to look for component sites for cheap USB dongles.

Those looking to exchange their old Pi with the new one will need to upgrade their power source, as the Pi 3 requires a 2.5A input. The Pi 3 can be bought from Element14, RS Components and the other usual stockists for US\$ 35. It is also available at <http://kitsnspares.com> for ₹ 3300.

Open source adoption by the Indian government

The government's decision to switch over to open source options which are free could result in substantial cost savings but is also a matter of concern in terms of security and operational efficiency, both of which need to be addressed.

The government's total IT spend was going to be US\$ 66.98 billion in 2015 as per the Ministry of Communications and IT. This was derived from several global IT research and advisory forecasts. According to NASSCOM's Data Security Council of India, the average expense on cyber security is nearly 2 to 3 per cent of the total IT spending, i.e., US\$ 1.5 to US\$ 2 billion.

As per Rahul De, Hewlett-Packard's chair professor on ICT for sustainable economic development at IIM Bangalore, a major proportion—almost 15-20 per cent—of the government's IT spend is on closed software. Transitioning

NGA releases seven tools under open source licence

The National Geospatial-Intelligence Agency (NGA) of the USA provides combat and intelligence agencies with critical and often classified intelligence. NGA is trying to share and collaborate on geodata in the open, and has expanded its transparency efforts with the release of seven open source tools on GitHub.

These are:

1. **DigitalGlobe Reader:** This is a C++ class loading TIFF imagery and associated XML metadata files, along with images obtained from Digital Globe's World View 3 satellite.
2. **SWIR signal detection:** This tool runs a spectral signature comparison between the provided spectral library file and a signature provided by the user.
3. **Social media picture explorer:** This, along with its user interface, utilises machine learning techniques to assist users in exploring social media by combining similar images and offering instant object recognition.
4. **Nounalyzer:** This tool categorises the nouns and entities in a RSS feed and displays the results through visualisations for analysts to distil information faster.
5. **WordPress revision slider:** This tool modifies a WordPress theme to interactively show a post's revisions and get readers comfortable with consuming stories written with version control.
6. **Rational polynomial coefficients mapper:** This tool maps an object's latitude, longitude and altitude to a two-dimensional point.
7. **Spectral library reader:** This reads files in the splib06a spectral reflectance database so that users can graph out the spectral reflectances of the materials in a multi-band image.

The GEOINT Pathfinder project was launched in August 2015, and assists the NGA in navigating the competitive world of commercial geospatial intelligence in an unclassified setting. According to sources in the NGA, the project endeavours to answer research questions via unclassified tools, data, information technology and services available from public or open sources.